

TTL CIRCUIT FAMILIES

TTL CMOS characteristics

ARN

Version 2016, β

- Each family is designed around certain performance objectives, within the economic limitations of a particular process.
- The key performance factors that distinguish the families are
 - power consumption,
 - speed and
 - ability to drive wiring capacitance.

TTL CIRCUIT FAMILIES cont.

- In three of the families - TTL, H-TTL and LP-TTL - the transistors are turned on by applying sufficient base current for the lowest expected current gain.
- The average transistor, having greater current gain, receives far more base current than necessary, which forward biases the collector-base junction and saturates the transistor.
- In order to turn off such a saturated transistor, the excess base charge must first be removed, resulting in considerable delay.
- Gold doping is commonly used to speed up the charge recombination, but this decreases the current gain.

TTL CIRCUIT FAMILIES cont.

- Schottky clamped transistors overcome this limitation.
- They use a **surface barrier diode** with very low forward voltage drop (0.3 V) as a bypass between base and collector.
- When the transistor starts conducting and is about to become saturated, the excess input current is not fed into the base, but routed through the Schottky diode into the collector (Baker Clamp). As a result the transistor is never fully saturated and recovers quickly when the base current is interrupted.
- Since gold doping is not required, the transistors also have higher current gain, require less base current, and turn on faster.

TTL CIRCUIT FAMILIES cont.

- As a result of the faster turn-on and recovery, S-TTL circuits achieve roughly twice the speed of H-TTL at about the same level of power consumption, as indicated in the table.
- On the other hand, LS-TTL circuits (also Schottky clamped) use much less power than H-TTL, yet operate at about the same speed.
- Compared to S-TTL, LS-TTL processing produces shallower diffusions and smaller transistors with greater bandwidth.
- Thus, LSTTL circuits operate at about half the speed of S-TTL while using only about 20% as much power.

TTL CIRCUIT FAMILIES cont

CIRCUIT TYPE		TTL	H-TTL	S-TTL	LP-TTL	LS-TTL
2-input NAND 7400	t_{PLH}/t_{PHL}	22/15	10/10	4.5/5.0		10/10
	I_{CC}	2.0/5.5	4.2/10	4.0/9.0		0.4/1.1
D-Type Flip-Flop 7474	t_{max}	15	35	75		30
	I_{CC}	7.0	38	25		4.0

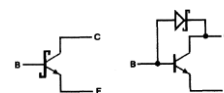


Fig. 2-1 Schottky Transistor

LP TTL gate schematic

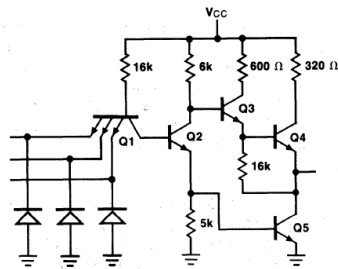


Fig. 2-4 LP-TTL Gate

TTL input characteristics

- With all inputs HIGH, the gate current flows through the base-collector junction of Q1 and turns on Q2.
- In this situation, a small quantity of charge is injected into the base of Q1. Part of this charge recombines in the base region and part of it drifts over to be "collected" by the emitters.
- This inverse beta current is a significant part of the input leakage current I_{IH} .
- This same phenomenon occurs when the gate current exits through a LOW input.
- Current is injected into the base from the LOW emitter and part of it is collected by the HIGH emitters.

TTL input characteristics cont.

- An input signal exceeding the +5.5 V rating applied to one input can cause breakdown between it and a LOW input, with the possibility of damage or of being biased in a negative resistance region, depending on the source impedance.
- Biasing in the negative resistance region can lead to oscillation that is difficult to diagnose.

Comparison LP v/s H TTL.

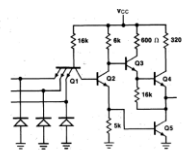


Fig. 2-4 LP-TTL Gate

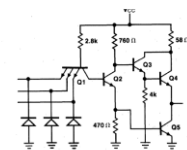
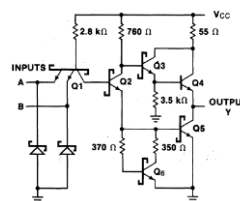


Fig. 2-5 74H10 Gate

- Several resistor values are halved and the output pull-up changed to a Darlington configuration.
- As seen from the values listed in the table, both speed and power consumption are approximately doubled

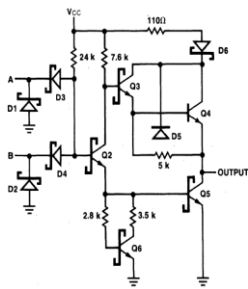
74S TTL

- The S-TTL gate is quite similar to the H-TTL gate and consumes about the same amount of power, yet operates at twice the speed.
- The base of the pull-down output transistor Q5 is returned to ground through Q6 and a pair of resistors instead of through a simple resistor.



- This arrangement is called a squaring network since it squares up the transfer characteristics by preventing conduction in the phase splitter Q2 until the input voltage rises high enough to allow Q2 to supply base current to Q5.
- The squaring network also improves the propagation delay by providing a low resistance path to discharge capacitance at the base of Q5 during turn-off.

74 LS TTL



- With a few exceptions, LS-TTL circuits do not use the multi-emitter input structure that originally gave TTL its name.
- Most LS elements use a DTL type input circuit with Schottky diodes to perform the AND function, as exemplified by Q3 and Q4 in Figure 2-7.
- Compared to the classical multi-emitter structure, this circuit is faster and it increases the input breakdown voltage.
- Inputs of this type are tested for leakage with an applied input voltage of 10V, and the input breakdown voltage is guaranteed to be 15 V or more

74 LS TTL

- the sneak diode D5 does not return to Vcc as it does in other diffused resistors. This feature, in conjunction with the blocking diode D6, allows a **HIGH state output to be pulled up higher than Vcc, e.g., to +10 V, convenient for interfacing with CMOS.**

INPUT CHARACTERISTICS

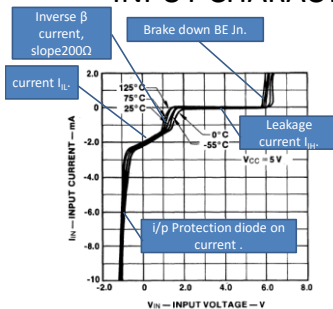


Fig. 2-10 TTL Buffer Input Characteristics

- For input voltage levels around 6.0V the emitter inputs avalanche.
- In this condition an input has very low series resistance and the avalanche current increases rapidly with input voltage.
- To avoid permanent damage, the input signal source must have either a current limit of 5.0 mA or less, or the input voltage must be limited to 5.5 V

INPUT CHARACTERISTICS cont.

- For input voltages between 2.0 V and 5.5 V, the current flow is only the input leakage current I_{IH} , guaranteed not to exceed 40 μ A or 50 μ A, depending on the circuit family, for a single input.

INPUT CHARACTERISTICS cont.

- As the input voltage decreases below 2.0 V, current starts flowing from the input and increases rapidly as the voltage decreases.
- The slope of the characteristics in this region is only about 200 ohm, indicating that part of the current from the gate pull-up resistor is still flowing inward through the internal junctions of the circuit.
- This marks the transition region, since at some point the buffer pull-down output transistor will not have sufficient base current to stay on and the output will start to switch from LOW to HIGH.

INPUT CHARACTERISTICS cont.

- As the input voltage decreases from about 1.0 V to -0.5 V, the slope of the characteristic equals the gate pull-up resistor, in this case about 2 k ohm.
- As the input goes below about -0.7 V the current increases rapidly as the input clamping diode conducts.

LS TTL INPUT CHARACTERISTICS

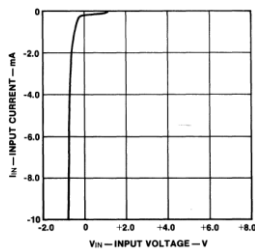


Fig. 2-11 LS-TTL Input Characteristics

- Input diode breakdown is typically greater than 15 V and input leakage current above 1.5 V is negligible.
- As the input voltage falls below 1.3 V, gate current starts flowing out of the input, denoting the transition region.
- For input voltage between 1.0V and -0.3V, the I-V characteristic has the slope of the 24 kohm gate pull-up resistor.

LS TTL INPUT CHARACTERISTICS

- The clamping diode conducts and the current increases rapidly when the input voltage goes below about -0.3V.
- The LS-TTL clamping diodes are intended only for the suppression of transient currents and should not be used as steady-state clamps in interface applications.
- A clamp current exceeding 2.0 mA and with a duration greater than 500 ns can activate a parasitic lateral npn transistor, which in turn can steal current from internal nodes of the LS circuit and thus cause logic errors.
- The effective capacitance of a TTL input is 5.0 pF for DIP and 4.0 pF for Flatpak.
- For an input that serves more than one internal function, each additional function adds 1.5 pF.

UNUSED INPUTS

- Theoretically, an unconnected input assumes the HIGH logic level, but practically speaking it is in an undefined logic state because it tends to act as an antenna for noise.
- Only a few hundred millivolts of noise causes the unconnected input to go to the logic LOW state.
- On devices with memory (flip-flops, latches, registers, counters), it is particularly important to terminate unused inputs (MR, PE, PL, CP) properly since a noise spike on these inputs might change the contents of the memory.
- It is poor design practice to leave unused inputs floating.

UNUSED INPUTS

- For a permanent HIGH signal, unused inputs can be tied to V_{CC} .
- A current limiting resistor, in the range of 1 K Ω to 5 K Ω , is recommended for emitter-type inputs since these break down at some unspecified voltage above 5.5 V and power supply mis adjustment or malfunction can cause damage unless the current is limited.
- Note that one resistor can serve several inputs, provided only that the cumulative I_{IH} current does not cause the voltage to drop below 2.4 V.

UNUSED INPUTS LS TTL

- Note also that diode-type LS-TTL inputs have breakdown voltages above 15 V and thus protective resistors are not normally required.
- An unused input may also be tied to a used input having the same logic function, such as NAND or AND gates, provided that the driver can handle the added I_{IH} .
 - This practice is not recommended for diode-type LS-TTL inputs in a noisy environment, since each diode represents a small capacitor and two or more in parallel can act as an entry port for negative spikes superimposed on a HIGH level and cause momentary turn-off of Q2

OUTPUT CHARACTERISTICS

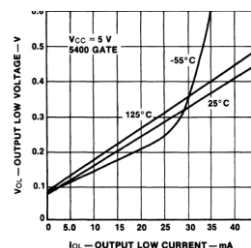
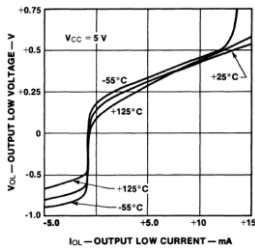


Fig. 2-12 TTL Gate Output LOW Characteristics

- With no load current the output level is the offset voltage of about 90 mV.
- The slope of the 25 $^{\circ}$ C characteristic indicates a saturation resistance of 8 ohm, increasing to 9 ohm at 125 $^{\circ}$ C and decreasing to 6 ohm at -55 $^{\circ}$ C.
- At low temperature, transistor beta decreases such that the output transistor pulls out of saturation for currents above 25 mA

OUTPUT CHARACTERISTICS V_{OL}



- In S-TTL and LS-TTL circuits, the anti-saturation clamping of Q5 means that the output LOW voltage will be slightly higher than for the equivalent saturated circuits at the same current levels.

OUTPUT CHARACTERISTICS V_{OL}

- For example, both the 74S00 and 74H00 are guaranteed to sink 20mA (12.5 unit loads) but the V_{OL} specifications are 0.5 V and 0.4 V, respectively.
- Similarly, 74LS circuits are guaranteed to sink 8.0 mA (5 unit loads) but the V_{OL} Specifications 0.5 V.

OUTPUT CHARACTERISTICS: V_{OH}

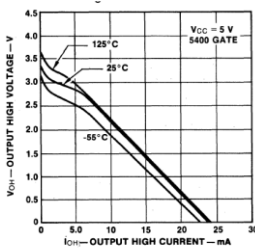


Fig. 2-14 TTL Output HIGH Characteristics

- With no load current the V_{OH} is about 3.5 V at 25°C.
- For I_{OH} increasing to about 6.0 mA, the characteristic has the shape of a fixed voltage minus the logarithmically increasing voltage drop across two pn junctions.
- For I_{OH} greater than 6.0mA, the pull-up transistor Q3 (Figure 2-2) saturates and the slope of the characteristic is just the 130 ohm the current limiting resistor plus the saturation resistance of Q3.

- The maximum I_{OH} current, where the characteristic intersects the horizontal axis, correlates with the short-circuit output current parameter I_{OS} and is often regarded as a measure of the circuit's ability to charge line capacitance.

LS TTL V_{OH}

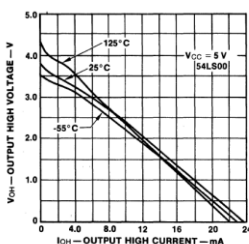


Fig. 2-16 LS-TTL Gate Output HIGH Characteristics

Due to the 5.0 k ohm resistor from Q3 to output (Figure 2-7) the LS circuits provide higher output voltage for low values of I_{OH} , as shown in Figure 2-17. This provides greater protection against negative-going noise on a quiescent HIGH signal

3-STATE OUTPUTS

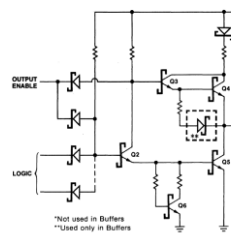


Fig. 2-18 Typical 3-State Output Control

- A LOW signal on the enable turns off both Q2 and Q3 and thus disables both the pull-up and pull-down circuitry.
- In this disabled condition the outputs are tested for leakage at 2.4 V (I_{OH}) and at 0.4 V or 0.5 V (I_{OL}) to ensure that they do not cause excessive loading on a data bus.

INCREASING FAN OUT

- When the circuit is enabled, the output HIGH and LOW characteristics are the same as those of other circuit types having the same drive capabilities

- To increase fan-out, inputs and outputs of gates on the same package may be paralleled.
- It is advisable to limit the gates being paralleled to those in a single package to avoid large transient supply currents due to different switching times of the gates.
- This is not harmful to the devices, but could cause logic problems if the gates are being used as clock drivers.

OPEN-COLLECTOR OUTPUTS

Voltage/ Current level comparisons

TABLE 8-9 Input/output voltage levels (in volts) with $V_{DD} = V_{CC} = +5V$.

Parameter	CMOS							TTL			
	4000B	74HC	74HCT	74AC	74ACT	74AHC	74AHCT	74	74LS	74AS	74ALS
$V_{IH}(\text{min})$	3.5	3.5	2.0	3.5	2.0	3.85	2.0	2.0	2.0	2.0	2.0
$V_{IL}(\text{max})$	1.5	1.0	0.8	1.5	0.8	1.65	0.8	0.8	0.8	0.8	0.8
$V_{OH}(\text{min})$	4.95	4.9	4.9	4.9	4.9	4.4	3.15	2.4	2.7	2.7	2.5
$V_{OL}(\text{max})$	0.05	0.1	0.1	0.1	0.1	0.44	0.1	0.4	0.5	0.5	0.5
V_{NH}	1.45	1.4	2.9	1.4	2.9	0.55	1.15	0.4	0.7	0.7	0.7
V_{NL}	1.45	0.9	0.7	1.4	0.7	1.21	0.7	0.4	0.3	0.3	0.4

TABLE 8-12 Input/output currents for standard devices with a supply voltage of 5V.

Parameter	CMOS				TTL					
	4000B	74HC/HCT	74AC/ACT	74AHC/AHCT	74	74LS	74AS	74ALS	74F	
$I_{IH}(\text{max})$	1 μA	1 μA	1 μA	1 μA	40 μA	20 μA	20 μA	20 μA	20 μA	
$I_{IL}(\text{max})$	1 μA	1 μA	1 μA	1 μA	1.6 mA	0.4 mA	0.5 mA	100 μA	0.6 mA	
$I_{OH}(\text{max})$	0.4 mA	4 mA	24 mA	8 mA	0.4 mA	0.4 mA	2 mA	400 mA	1.0 mA	
$I_{OL}(\text{max})$	0.4 mA	4 mA	24 mA	8 mA	16 mA	8 mA	20 mA	8 mA	20 mA	

CMOC 40xx series

- Devices in the 4000/14000 series have very low power dissipation and can operate over a wide range of power-supply voltages (3 to 15 V).
- They are very slow compared to TTL and other CMOS series and have very low output current capabilities.
- They are not pin-compatible or electrically compatible with any TTL series

74HC/HCT (High-Speed CMOS)

- The 74HC series has a 10-fold increase in switching speed, comparable to that of the 74LS devices, and a much higher output current capability than the first 7400 CMOS series ICs.
- 74HC/HCT ICs are pin-compatible with and functionally equivalent to TTL ICs with the same device number.

74AHC/AHCT (Advanced High-Speed CMOS)

- a natural migration path from the HC series to faster, lower-power, low-drive applications.
- The devices in this series are three times faster and can be used as direct replacements for HC series devices.
- They offer similar noise immunity to HC without the overshoot/undershoot problems often associated with higher drive characteristics required for comparable speed.